

Improving Game Based Learning Through AI-Driven Interactivity

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Abstract

Game-based learning (GBL) has demonstrated potential as an instructional approach across educational domains, yet the role of interactivity as a design variable remains underexplored. This study examined whether game-based and non-game-based formats produce significant knowledge gains, and whether integrating a large language model (LLM)-driven veterinary non-player character (NPC) into a game environment enhances learning outcomes and engagement beyond a static in-game reference. Using a between-subjects design, participants were assigned to one of three conditions: a non-game caresheet control, a game with a static in-game caresheet, and a game with an LLM-driven veterinary NPC. Knowledge was assessed via a multiple-choice quiz administered before and after the intervention, and learner engagement was measured using an adapted version of the Game Experience Questionnaire. Results indicated that all conditions produced significant knowledge gains overall, and that the magnitude of improvement differed significantly across conditions, though post hoc comparisons did not locate significant differences between specific condition pairs, consistent with limited statistical power. The Game/LLM-Interactive condition reported significantly higher engagement than the Non-Game Control. These findings suggest that embedding conversational AI into a game-based learning environment meaningfully enhances engagement compared to a non-game baseline.

Keywords: game-based learning, large language models, engagement, knowledge acquisition

1. Introduction

Understanding when and how games support knowledge acquisition has become an important question for educators and researchers, as digital tools become increasingly embedded in both formal and informal learning environments (Chiotaki et al., 2023). Game-Based Learning (GBL) has emerged as a widely studied instructional approach with documented potential across a range of educational domains (Plass et al., 2015; Talan et al., 2020). GBL refers to instructional activities that use games to achieve learning goals (Talan et al., 2020). Despite a growing body of evidence, it remains unclear under which conditions GBL outperforms traditional instruction. In particular, the role of interactivity as a design variable, including how much interactivity is needed to produce learning gains, has received limited attention in applied educational settings.

Furthermore, an underexplored domain within GBL is teaching leopard gecko husbandry to novice reptile owners. Reptile ownership has grown considerably in recent years (APPA, 2022), yet accessible, evidence-based care education remains scarce (Azevedo et al., 2021). Gaps in foundational knowledge about feeding, habitat setup, and behavioral recognition are a primary cause of preventable harm to captive reptiles (Azevedo et al., 2021; Howell & Bennett, 2017). A game-based format offers a promising way to close this gap by presenting species-specific knowledge in an engaging, repeatable structure.

Aside from GBL, recent advances in AI have also shown potential in an educational context. Large language models (LLMs) can now serve as interactive learning companions. They respond to open-ended questions, offer personalized explanations, and adapt to individual learners in ways that fixed reference materials cannot (Bernacki et al., 2024; Kasneci et al., 2023). In reptile owner-facing contexts, AI chatbots have shown promise in sharing validated care information, and acceptance of AI-based veterinary advice is growing, particularly when professional access is limited (Huang & Chueh, 2021; Jokar et al., 2024). In educational settings more broadly, LLMs can function as on-demand tutors. They answer follow-up questions, adjust explanations to the learner's level, and support active reasoning in ways that static text cannot (Kasneci et al., 2023; Bernacki et al., 2024). Combining these two trends, AI in veterinary education and LLMs as interactive learning tools, points naturally toward embedding a conversational AI veterinarian inside a learning game.

Therefore, two interconnected questions are being asked. First, do game-based and non-game-based formats each produce significant improvements in participants' knowledge of leopard gecko care? Second, within the game-based format, does access to an LLM-driven veterinary Non-Playable Character (NPC) produce stronger learning outcomes than access to a static in-game caresheet? This study compares three instructional conditions: a non-game caresheet (control), a game with a static in-game caresheet, and a game with an LLM-driven veterinary NPC. All three conditions cover the same learning content, derived from a validated reptile care source. Knowledge is measured before and after the intervention using a purpose-built quiz, and learner experience is assessed via questionnaire. The core research question is whether adding an LLM-driven conversational layer to a game-based format produces meaningfully stronger learning outcomes than the game alone.

2. Methods

2.1 Research Design

This study employed a between-subjects experimental design with three active experimental conditions and a pre-test/post-test measurement structure. Participants were assigned to one of three instructional formats varying in game presence and level of AI-driven interactivity. Condition assignment was distributed manually to ensure balanced group sizes. The design allowed for both within-condition learning assessments (comparing pre- to post-test scores within each group) and between-condition comparisons (examining whether different formats produced differential learning outcomes).

The four experimental conditions of the broader conceptual framework were defined by two dimensions: whether the learning material was delivered within a game environment, and whether an LLM-driven conversational assistant was available. These experimental conditions are summarized in Table 1. Of the four, Non-Game Control, Game / Non-LLM Interactive, and Game / LLM-Interactive were actively administered in the current study. Non-Game / LLM-Interactive was excluded, as the present study focused specifically on the contribution of LLM interactivity within a game-based context; Non-Game Control served as the sole non-game control.

Table 1*Overview of Experimental Conditions*

Condition	Format	Delivery Method
Non-Game Control	Traditional	Information provided via caresheet only (no game elements). Serves as the primary control condition.
Non-Game / LLM-Interactive	Traditional + AI	Information provided via caresheet with access to an LLM-driven conversational assistant. Not tested in the current study.
Game / Non-LLM Interactive	GBL	Information embedded in the game environment via an in-game caresheet; no AI NPC guidance.
Game / LLM-Interactive	GBL + AI	Game environment with a veterinary NPC powered by LLM/Chatbot logic and conversational interaction.

Note. NPC = Non-Player Character; LLM = Large Language Model. Condition 2 is defined within the conceptual framework but not administered in the current study.

Non-Game Control presented the learning content through a static caresheet without any game elements, serving as the traditional instruction baseline. Game / Non-LLM Interactive embedded the same caresheet content within the game interface, but provided no interactive NPC guidance. Game / LLM-Interactive replaced the static reference with a conversational LLM-driven veterinary NPC.

The learning content was consistent across all active conditions and was organized around two gameplay modules: Feeding and Terrarium Building. The Feeding module covered three concepts: that leopard geckos are strict insectivores, that dietary variety across feeder insect species was essential for nutritional balance, and that calcium and multivitamin supplementation prevents metabolic bone disease. The Terrarium Building module covered three corresponding concepts: appropriate substrate composition, the role of environmental enrichment in supporting natural behaviors, and the life support requirements for a healthy enclosure, including heating, UVB lighting, and humidity management. All learning content was derived from a widely used caresheet (ReptiFiles, 2026).

Furthermore, the NPC was friendly and approachable in tone, and was designed to guide players through the learning content rather than simply providing answers. Players could ask open-ended questions in natural language. The NPC was constrained by its system prompt to only discuss topics related to leopard gecko husbandry as covered in the caresheet, and would not engage with off-topic queries. To keep interactions focused, players were limited to 10 questions per module.

2.2 Participants and Sampling Strategy

Participants were recruited and tested in person. To be eligible, participants had to be proficient in English and comfortable using modern technology, specifically a laptop and mouse, as the game interface and all study materials were digital and text-based. In order to establish the required sample size, prior GBL based studies were analyzed. Results of this analysis can be found in Table 2.

Table 2

Comparison of Results Across Recent GBL Studies

Topic	Sample (N)	Primary Finding	Statistical Result
Dental Trauma	88	GBL = Traditional Lecture	Not Significant ($p > .05$)
Pharmacology	98	GBL > Tutorials	Highly Significant ($p < .001$)
Mathematics	29	Significant Improvement	Highly Significant ($p < .05$)
Volleyball Skills	36	GBL = Direct Instruction	Not Significant ($p > .05$)
Microbiology	30	GBL = Problem-Based Learning	Not Significant ($p > .05$)

Note. GBL = Game-Based Learning; PBL = Problem-Based Learning.

Sample size was estimated at 20 participants per condition (N = 60 total), based on the sample sizes observed in the prior studies reviewed in Table 2. The smallest study in that set (N = 29) successfully detected significant within-condition learning gains, while studies examining between-condition differences operated with samples ranging from 30 to 98. A target of 20 per condition was considered feasible for in-person recruitment and sufficient to detect within-condition pre-to-post improvement. The study is therefore exploratory with respect to between-condition comparisons. Condition assignment was performed to ensure balanced group sizes across conditions.

2.3 Materials

The leopard gecko caretaking game was developed specifically for this study. All conditions drew on the same factual knowledge base: the ReptiFiles Leopard Gecko Care Sheet authored by Mariah Healey (ReptiFiles, 2026). Written consent was obtained from Healey to use this material as the informational foundation for the game conditions and as the caresheet presented in the Non-Game Control condition.

The game consisted of two interactive modules, Feeding and Terrarium Building, each structurally identical across the three game conditions. In the Feeding module, players prepared a meal for the gecko by dragging ingredients into a food bowl. In the Terrarium Building module, players assembled the gecko's enclosure by selecting and placing components. The task did not change across conditions; what differed was how players could access information to guide their decisions.

In the Game / Non-LLM Interactive condition, an in-game caresheet was available within the game interface, presenting the relevant husbandry information as a readable reference. In the Game / LLM-Interactive condition, the veterinary NPC was powered by Google Gemini 3.1 Flash Lite, configured via a domain-specific system prompt constraining responses to leopard gecko husbandry as described in the ReptiFiles caresheet. Players interacted with the NPC through a chat interface embedded in the game. Screenshots of both the in-game caresheet interface and the LLM veterinarian chat interface are provided in Figure 1.



Figure 1

Comparison of in-game caresheet and in-game LLM veterinarian

Note. Left: in-game caresheet; Right: in-game LLM veterinarian.

Knowledge was assessed using a purpose-built 10-item multiple-choice assessment, administered before (pre-test) and after (post-test) the intervention. Items covered the six learning concepts targeted by the two modules, plus one additional control item that assessed factual knowledge about leopard gecko shedding frequency. This control item tested information that was not taught in any condition, and served as a within-instrument consistency check rather than a learning outcome measure. An example question was: *Which of the following describes the natural diet of a Leopard Gecko?*. All items included a scored distractor set and an explicit 'I don't know' option to reduce guessing. Pre- and post-test forms used identical items in a different order to reduce practice effects. The full assessment is reproduced in Appendix A.

Learner satisfaction and perceived engagement were assessed using an adapted version of the In-Game Game Experience Questionnaire (GEQ: IJsselsteijn et al., 2013), administered as part of the post-survey. The GEQ captured dimensions including immersion, competence, flow, and perceived learning value. The questionnaire was modified to avoid the word 'game' so that it applied equally to all conditions, including the Non-Game Control. The full GEQ instrument is included in Appendix B.

Knowledge gain was operationalized as the difference between post-test and pre-test scores, excluding the control item. Success was defined as a statistically significant improvement in post-test scores ($p < .05$). Between-condition differences were examined using a mixed factorial ANOVA on improvement on pre- vs. post-test scores, as well as engagement scores following the GEQ being examined by a one-way ANOVA.

2.5 Procedure

A total of 100 participants were recruited across the full study, which included an additional condition tested in a parallel study and excluded from the present analyses. Participants were recruited and completed the study in person while being monitored by a researcher. Upon arrival, participants provided informed consent and supplied their first name, which was recorded solely for age verification purposes (participants had to be 16 or older). Participants then completed the pre-survey knowledge quiz. They were manually assigned to one of the three conditions and directed to the corresponding instructional material, following an alternating rotation. After completing the intervention, participants completed the post-survey, which included the equivalent knowledge quiz and the adapted GEQ. Upon finishing, participants received a printed information sheet summarizing the study's purpose and providing contact details for follow-up questions. The estimated total duration per participant was 20 to 30 minutes.

2.6 Ethical Considerations

This study involved human participants and was designed in accordance with standard ethical principles for educational research. Participants were only tested with when aged above 16. No personal data was gathered from participants aged 16-17. Participation was voluntary, and participants could withdraw at any time without consequence. Informed consent was obtained from all participants prior to the start of the study. The only personal information collected was a participant's name, retained solely for data identification, should participants have wanted to get their data revoked from the study. Participants were informed of the study's purpose and received a printed debrief sheet upon completion. The study did not involve deception, sensitive personal topics, or any procedures that could cause distress. The LLM-driven veterinary NPC was constrained to discuss only leopard gecko husbandry topics; this boundary was enforced via the system prompt to prevent exposure to unintended or off-topic AI-generated content.

3 Results

3.1 Participant Exclusions and Final Sample

Prior to analysis, pre- and post-test knowledge scores were visually inspected to identify potential outliers. Two cases with pre-test scores of 7 were excluded, as these fell substantially above the remainder of the distribution (range: 0–6). One case with a post-test score of 1 was similarly excluded as a low outlier relative to the main cluster (range: 2–9). Following exclusions, the final analytic sample comprised 70 participants distributed across three conditions: Game/Non-LLM Interactive ($n = 23$), Game/LLM-Interactive ($n = 20$), and Non-Game Control ($n = 27$).

3.2 Knowledge Scores

A mixed factorial repeated-measures ANOVA was conducted to examine changes in knowledge scores from pre- to post-test across conditions. Descriptive statistics and inferential results are reported in Table 3. A significant main effect of time was observed, $F(1, 67) = 151.85, p < .001$, indicating that knowledge scores improved from pre- to post-test over conditions. A significant Time $\times$ Condition interaction was also found, $F(2, 67) = 3.52, p < .05$, indicating that the magnitude of knowledge gain differed across conditions. However, post hoc Bonferroni-corrected pairwise comparisons and simple contrasts did not reveal significant differences between any specific condition pairs, all $p > .05$. Mean knowledge gain within this sample was highest in the Game/Non-LLM Interactive condition, $M = 3.74, SD = 2.62$, followed by the Game/LLM-Interactive condition, $M = 3.25, SD = 1.94$, and lowest in the Non-Game Control condition, $M = 2.22, SD = 1.58$.

Table 3

Descriptive Statistics and Repeated Measures ANOVA Results for Knowledge Scores by Condition

	Game/Non-LLM Interactive (n = 23)	Game/LLM- Interactive (n = 20)	Non-Game Control (n = 27)	<i>F</i>	<i>df</i>	<i>p</i>
Pre-test M (SD)	2.65 (2.01)	3.00 (2.08)	2.93 (1.88)			
Post-test M (SD)	6.39 (1.44)	6.25 (1.52)	5.15 (1.49)			
Knowledge Gain M (SD)	3.74 (2.62)	3.25 (1.94)	2.22 (1.58)			
Time effect				151.85	1, 67	< .001
Time × Condition				3.52	2, 67	< .05
Between-subjects (Condition)				1.21	2, 67	> .05
Note. Pre- and post-test scores reflect raw knowledge scores. Knowledge gain reflects the difference between post- and pre-test scores.						

3.3 Engagement Scores

A one-way ANOVA revealed a significant effect of condition on engagement scores, $F(2, 67) = 4.32, p < .05, R^2 = .114$; see Table 4. The Game/LLM-Interactive condition reported the highest mean engagement, $M = 51.20, SD = 4.61$, followed by the Game/Non-LLM Interactive condition, $M = 49.52, SD = 6.13$ and the Non-Game Control condition, $M = 46.07, SD = 7.08$. Post hoc Bonferroni comparisons indicated that the Game/LLM-Interactive condition produced significantly higher engagement than the Non-Game Control condition, mean difference = 5.13, $SE = 1.81, p < .05$, a result further supported by simple contrasts, estimate = 5.13, $p < .05$. No significant difference was found between the two game conditions, $p > .05$ or between the Game/Non-LLM Interactive and Non-Game Control conditions, $p > .05$.

Table 4*Descriptive Statistics and One-Way ANOVA Results for Engagement Scores by Condition*

	Game/Non-LLM Interactive (n = 23)	Game/LLM-Interactive (n = 20)	Non-Game Control (n = 27)	<i>F</i>	<i>df</i>	<i>p</i>
Engagement M (SD)	49.52 (6.13)	51.20 (4.61)	46.07 (7.08)	4.32	2, 67	< .05

Pairwise Comparisons (Bonferroni)

Comparison	Mean Difference	<i>SE</i>	<i>p</i>
Game/Non-LLM Interactive vs. Game/LLM-Interactive	-1.68	1.88	> .05
Game/Non-LLM Interactive vs. Non-Game Control	3.45	1.75	> .05
Game/LLM-Interactive vs. Non-Game Control	5.13*	1.81	< .05

Note. Pairwise comparisons use Bonferroni correction. $R^2 = .114$. * $p < .05$.

4. Discussion

This study investigated whether AI-driven interactivity in a serious game enhances knowledge acquisition and engagement in leopard gecko husbandry education. A between-subjects design compared three conditions, a non-game caresheet control, a game with static in-game information, and a game featuring an LLM-driven veterinary NPC. Across 70 participants, knowledge gain was measured through a pre- and post-test quiz and engagement through an adapted Game Experience Questionnaire.

All three conditions combined, the Game / Non-LLM Interactive Condition, the Game / LLM-Interactive Condition, and the Non-Game Control Condition, successfully improved participant knowledge from pre- to post-test with the overall improvement being highly significant. This finding is consistent with the broader GBL literature, in which both game-based and traditional formats have been shown to support knowledge acquisition across a range of educational domains (Plass et al., 2015; Talan et al., 2020). Importantly, the magnitude of this knowledge difference over time differed significantly across conditions, but this significant difference could not be located in specific condition pairs through contrasts and post hoc testing.

With respect to engagement, the Game / LLM-Interactive Condition generated significantly higher engagement than the Non-Game Control Condition, while no significant difference was found between the two game conditions. This aligns with the view that LLM-driven conversational agents can meaningfully enhance the learner experience beyond passive reference formats (Bernacki et al., 2024; Kasneci et al., 2023), and extends this to a game-based context in which the NPC serves as an interactive learning companion.

Several limitations constrain the conclusions that can be drawn from this study. First, the sample size was modest. With 20-27 participants per condition, the study was adequately powered to detect within-condition pre-to-post improvements but underpowered with respect to between-condition comparisons. The significant time $\times$ condition interaction effect, paired with the absence of any significant pairwise differences, is consistent with an underpowered design rather than true equivalence across conditions. The numerical pattern of knowledge gain, highest in the Game / Non-LLM Interactive Condition, moderate in the Game / LLM-Interactive Condition, and lowest in the Non-Game Control, may reflect a genuine differential effect that a larger sample would resolve.

Furthermore, the participant pool was recruited in person and constrained by practical availability, which introduces self-selection concerns. Participants who agreed to take part in an in-person study on reptile care education may differ systematically from the broader population of novice reptile owners, particularly in baseline curiosity or technological comfort.

Following up, the study design did not include a delayed retention test. All post-test knowledge measurement occurred immediately after the intervention, meaning it is unknown whether the observed gains persisted over time.

Finally, the exclusion of the Non-Game / LLM-Interactive condition, while justified by the study's focus on GBL, means that the independent contribution of the LLM component cannot be isolated from the contribution of the game environment itself. The significantly higher engagement observed in the Game / LLM-Interactive Condition relative to the Non-Game Control could reflect the game, the LLM, or their combination. Tackling these effects would require the full 2×2 factorial design.

All in all, this study showed that all conditions significantly improved participant knowledge from pre- to post-test within a single session. Furthermore, the magnitude of this knowledge difference over time differed significantly across conditions, however the difference could not be located. This means that LLM-driven interactivity did not produce a significant advantage in learning outcomes. Whether that advantage emerges over longer exposure, remains an open question .

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Appendix A

Knowledge Assessment — Pre-Test and Post-Test Items

The following table lists all ten items used in both the pre-test and post-test. Items are presented in the same order in both forms; presentation order within the form differs between pre- and post-test to reduce practice effects. Correct answers are marked with an asterisk (*). Item 6 is a control item: the information it tests is not taught in any condition and is excluded from the primary knowledge gain analysis.

#	Question	Answer options
1	Which of the following describes the natural diet of a Leopard Gecko?	a) A variety of fruits and insects b) Exclusively insects * c) A mix of leafy greens and insects d) Small rodents and vegetables e) I don't know
2	What is the primary reason for meal variety?	a) To prevent the gecko from becoming overweight b) To ensure easy digestion c) To increase the gecko's water intake d) To provide a wider range of nutrients * e) I don't know
3	What is the secondary reason for meal variety?	a) To provide mental stimulation to the gecko * b) To encourage daytime activity c) To help the gecko shed skin more easily d) To improve the gecko's vision e) I don't know

4	What medical condition is most likely to develop if a Leopard Gecko lacks sufficient calcium?	a) Metabolic Bone Disease * b) Cellulose Toxicity c) Respiratory Infection d) Impaction e) I don't know
5	What are the components of the recommended substrate for Leopard Gecko enclosures?	a) Pure calcium sand b) Paper towels and ceramic tiles c) Bark, Moss and river stones d) Topsoil, sand and clay * e) I don't know
6 (Control)	How often do adult Leopard Geckos typically shed their skin?	a) Every 1 to 4 weeks * b) Every 4 to 8 weeks c) Every 8 to 12 weeks d) Every 12 to 16 weeks e) I don't know
7	Which of these is the recommended placement for a heat lamp?	a) In the middle of the enclosure for even heat b) Directly over the water dish c) Directly over the basking stone * d) Inside a cave e) I don't know
8	What is the function of providing a cave?	a) To make the terrarium look nicer b) To help the gecko catch prey more easily c) To provide a secure hiding spot *

		d) To provide a spot for the gecko to digest food e) I don't know
9	What is the recommended frequency for using supplements with young geckos?	a) Every feeding session * b) Once every three feeding sessions c) Once a week d) Only when the gecko is shedding e) I don't know
10	In the terrarium, what is the primary function of placing a flat stone?	a) To help with the gecko's heat regulation * b) To serve as a hunting point c) To absorb excess moisture from the air d) To provide nutrients to the gecko e) I don't know

Note. Correct answers are indicated with an asterisk (*). Item 6 is a control item not included in the primary learning outcome analysis.

Appendix B

Adapted Game Experience Questionnaire (GEQ) — Post-Survey Items

The following 14 items are drawn from the In-Game module of the Game Experience Questionnaire (IJsselstein et al., 2013). The word 'game' has been removed from all items so that the questionnaire applies equally across all conditions. Participants rate each item on a five-point scale: not at all, slightly, moderately, fairly, extremely.

1. I was interested in the knowledge presented
2. I felt successful
3. I felt bored
4. I found it impressive
5. I forgot everything around me
6. I felt frustrated
7. I found it tiresome
8. I felt irritable
9. I felt skillful
10. I felt completely absorbed
11. I felt content
12. I felt challenged
13. I had to put a lot of effort into it
14. I felt good